🎬 **MOVIEIQ**

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🎛️ Data Filters

🎭 Filter by Genre

All

⭐ Minimum Vote Average

0.00

MOVIEIQ

📊 Key Performance Indicators

Get a quick overview of the MovieIQ dataset and explore key movie performance metrics.

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Explore movie performance, discover genre trends, analyze box office economics, and predict potential movie revenue using Machine Learning.

🎬 Total Movies	💰 Average Re…	⭐ Average Ra…	📈 Average Pr…
2,000	$17…	6.04	$76,…

📋 Movie Dataset

📋 Movie Dataset

Showing 2,000 movies based on the selected filters.

	budget	revenue	popularity	runtime	vote_average	title	genres
0	57755036	107323371	60.3774	98	6.6119	Movie 1	[object Objec
1	192100010	342896533	86.2899	96	7.1847	Movie 2	[object Objec
2	128521863	253206827	86.9953	148	3.543	Movie 3	[object Objec
3	156299516	89635602	72.7101	118	5.9852	Movie 4	[object Objec
4	148532820	221133868	74.8296	161	4.5785	Movie 5	[object Objec
5	134224038	270524761	56.2348	150	7.3463	Movie 6	[object Objec
6	36788921	73368473	58.8821	160	4.1373	Movie 7	[object Objec
7	134120466	332939335	14.2314	118	8.3534	Movie 8	[object Objec
8	161953558	346179511	44.9421	115	5.5856	Movie 9	[object Objec
9	94410762	281375537	45.6269	119	4.188	Movie 10	[object Objec

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🔍 Data Quality Analysis

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All

⭐ Minimum Vote Average

0.00

🔍 Data Quality Analysis

Before building the machine learning model, we check the dataset for missing values and zero values in budget and revenue.

📋 Missing Values

	Column	Missing Values
0	budget	0
1	revenue	0
2	popularity	0
3	runtime	0
4	vote_average	0
5	title	0
6	genres	0
7	profit	0
8	success	0
9	genre	0

💰 Movies with Zero Budget

0

🎬 Movies with Zero Revenue

0

A budget or revenue value of zero may represent missing or unavailable financial information rather than a genuinely zero value. Such records can affect profit calculations and the accuracy of machine learning models. These rows should be reviewed and handled appropriately before final model training.

🎯 Movie Success Analysis

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All

⭐ Minimum Vote Average
0.00

🔥 Top 10 Most Popular Movies

🎬 Movie Performance Overview

🔥 Highest Popularity	💰 Highest Revenue	📈 Highest Profit
99.76	$574,49…	$381,39…

🔥 Top 10 Most Popular Movies

💰 Top 10 Most Profitable Movies

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All

⭐ Minimum Vote Average

0.00

💰 Top 10 Most Profitable Movies

📊 Success Factor Analysis

This analysis compares successful and unsuccessful movies based on popularity, runtime, and average audience rating.

📈 Average Performance by Movie Success

	popularity	runtime	vote_average
Unsuccessful	47.4041	131.6166	6.1192
Successful	50.7333	129.0781	6.017

🎬 Success Factor Comparison

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🎭 Filter by Genre

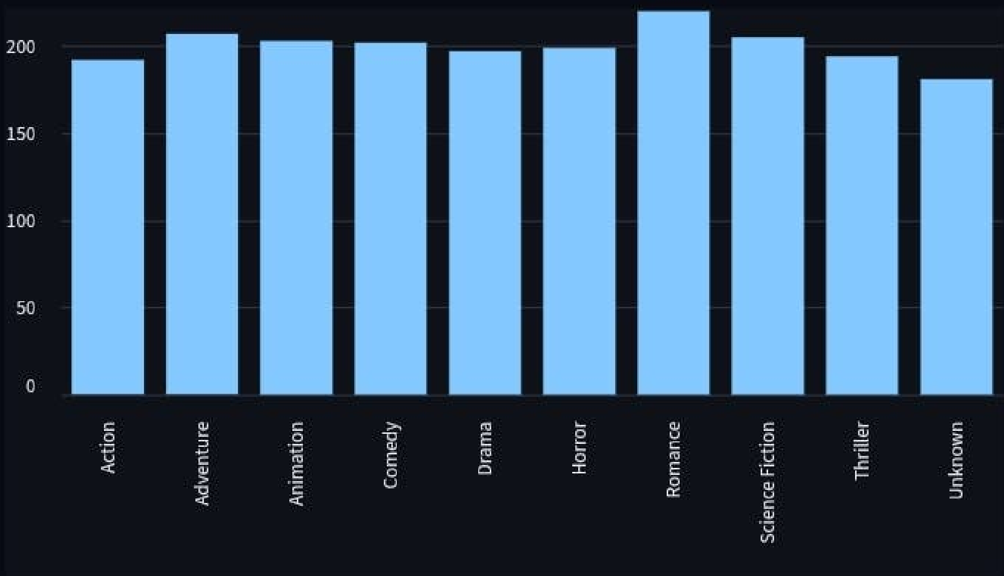
All

⭐ Minimum Vote Average
0.00

🎭 Genre Lab

Explore movie performance by genre. Select a genre below to discover its financial and audience insights.

🎭 Movies by Genre



🎬 Select a Genre

Action

📊 Action Insights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📊 Action Insights
🎬 Movies
192
💰 Avg Revenue
$18...
📈 Avg Profit
$79,...
⭐ Avg Rating
6.10
💰 Revenue vs Profit
150,000,000
100,000,000
50,000,000
0
Average Profit
Average Revenue
🎬 Movies in Action


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All
⭐ Minimum Vote Average
0.00
💰 Financial Analysis
Explore the financial performance of movies and understand the relationship between production budget and box office revenue.
🎬 Avg Budget
$10...
💵 Avg Revenue
$17...
📈 Avg Profit
$76,...
🔗 Budget-Re...
0.76
📈 Financial Performance Overview
180,000,000
160,000,000
140,000,000
120,000,000
100,000,000
80,000,000
60,000,000
40,000,000
20,000,000
0
Average Budget
Average Profit
Average Revenue
📊 Budget vs Revenue

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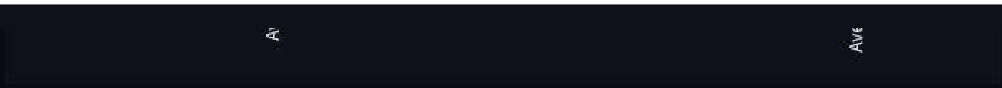
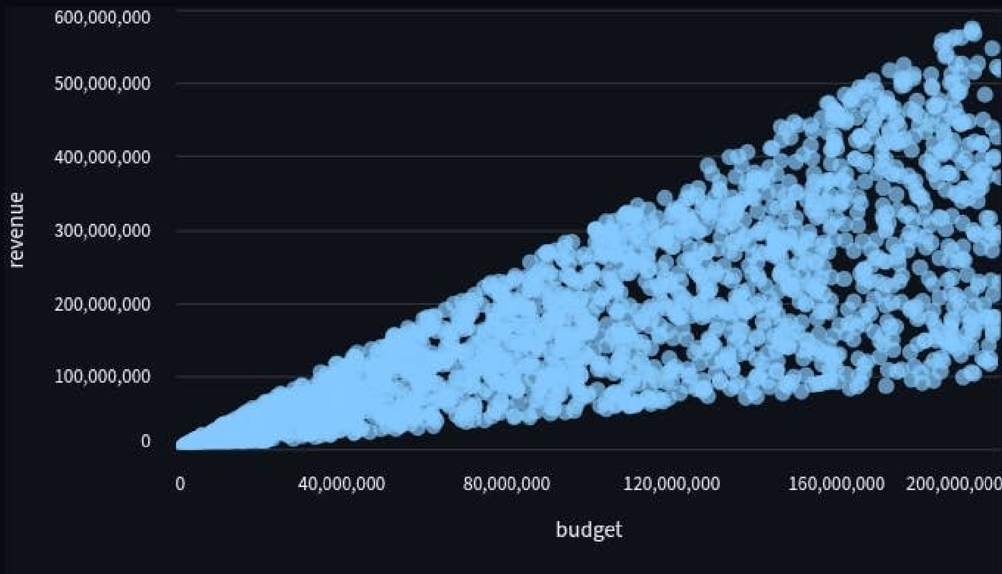
All

⭐ Minimum Vote Average
0.00



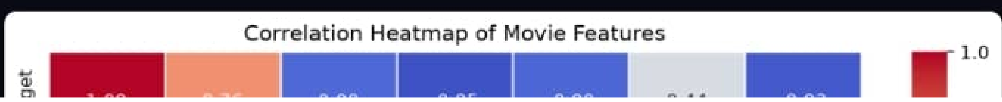
📊 Budget vs Revenue

Each point represents a movie. The chart helps us understand whether movies with larger budgets tend to generate higher revenue.



🔥 Correlation Heatmap

The correlation heatmap shows the strength of relationships between the main numeric movie features. Values closer to 1 indicate a strong positive relationship, values closer to -1 indicate a strong negative relationship, and values near 0 indicate a weak relationship.



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🎭 Filter by Genre

All

⭐ Minimum Vote Average
0.00

🔥 Correlation Heatmap

The correlation heatmap shows the strength of relationships between the main numeric movie features. Values closer to 1 indicate a strong positive relationship, values closer to -1 indicate a strong negative relationship, and values near 0 indicate a weak relationship.

The heatmap helps identify which numeric features are strongly related to each other. Strong correlations should be considered when interpreting the machine learning model, as highly related features may provide overlapping information.

💡 Key Insight

The budget-revenue correlation is 0.76. This indicates a positive relationship between movie budget and revenue in this dataset. However, correlation does not mean that budget alone determines a movie's success.

Statistical tests are used to identify whether relationships and differences observed in the movie dataset are statistically significant.

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⭐ Minimum Vote Average

0.00

🤩 Movie Success Predictor

Predict whether a movie is likely to be financially successful based on its budget, popularity, runtime, and expected audience rating.

MovieIQ uses a Random Forest Classification model trained on historical movie data to predict whether a movie is likely to generate revenue greater than its production budget.

🌟 Feature Importance

Feature importance shows which input variables contribute most to the Random Forest model's prediction of movie success.

📋 Feature Importance Table

	Feature	Importance
0	budget	0.2631
1	popularity	0.2631
3	vote_average	0.262
2	runtime	0.2117

📊 Feature Importance Visualization

2	runtime	0.2117

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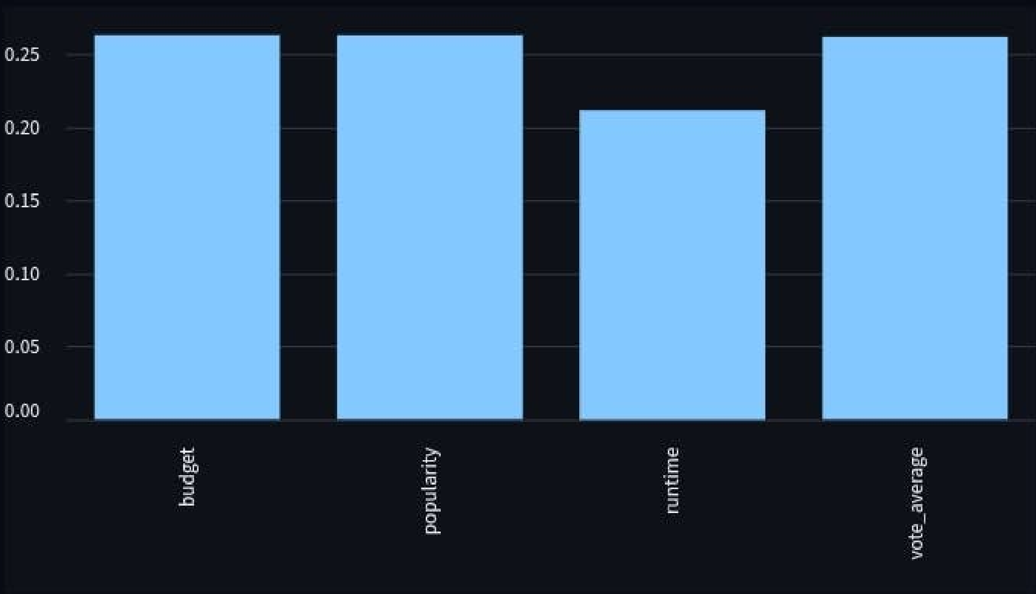
All

⭐ Minimum Vote Average
0.00

📊 Feature Importance Visualization



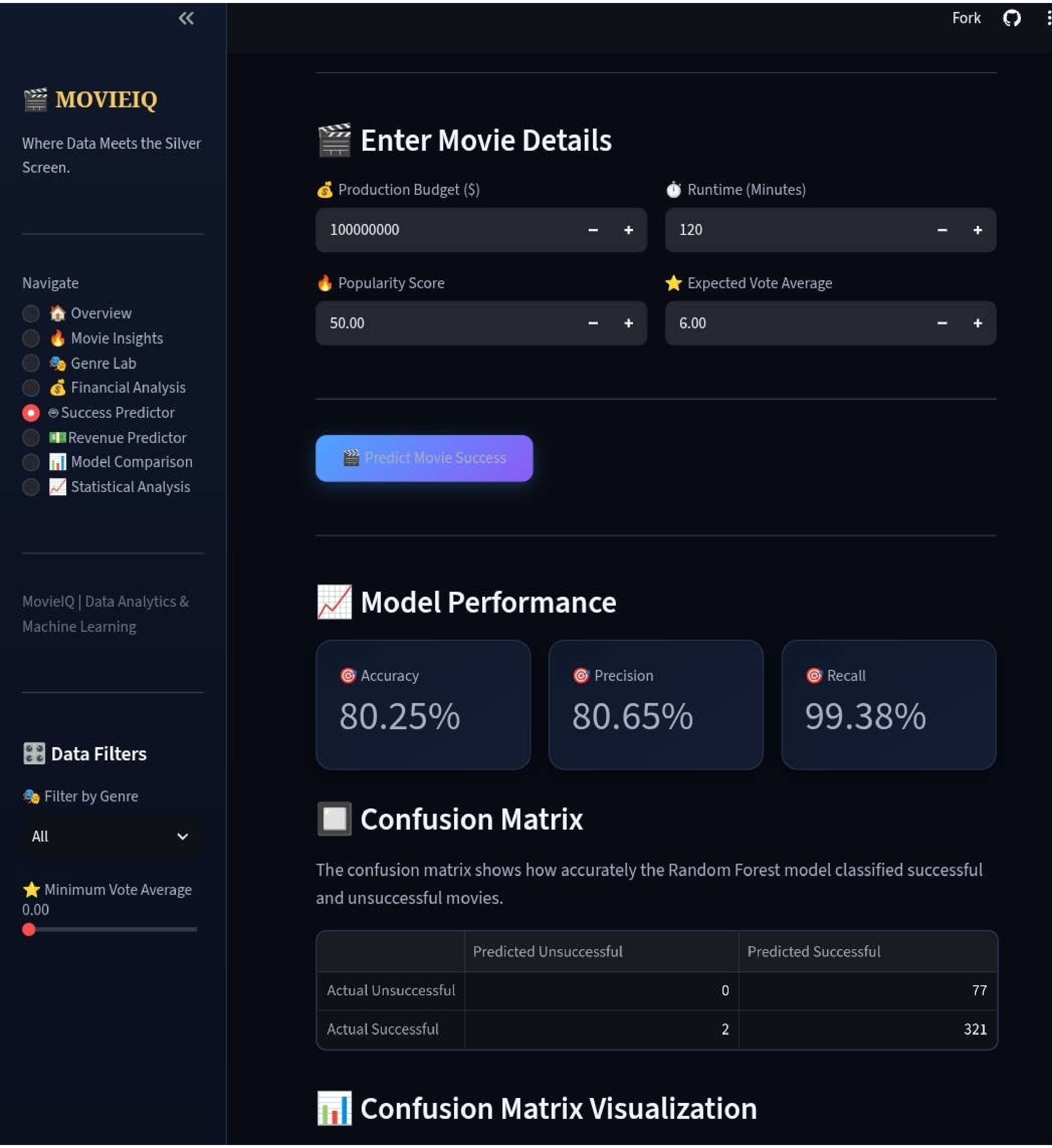
🌟 The most important feature for predicting movie success in the Random Forest model is **budget**, with an importance score of 0.2631.

Feature importance indicates how useful each feature was to the Random Forest model when making predictions. A higher importance score means the feature contributed more to the model's decisions. However, feature importance does not prove that a feature directly causes movie success.

🎬 Enter Movie Details

💰 Production Budget ($)

⏱️ Runtime (Minutes)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All
Minimum Vote Average
0.00
Enter Movie Details
Production Budget ($)
100000000
Runtime (Minutes)
120
Popularity Score
50.00
Expected Vote Average
6.00
Predict Movie Success
Model Performance
Accuracy
80.25%
Precision
80.65%
Recall
99.38%
Confusion Matrix
The confusion matrix shows how accurately the Random Forest model classified successful and unsuccessful movies.
Predicted Unsuccessful
Predicted Successful
Actual Unsuccessful
0
77
Actual Successful
2
321
Confusion Matrix Visualization

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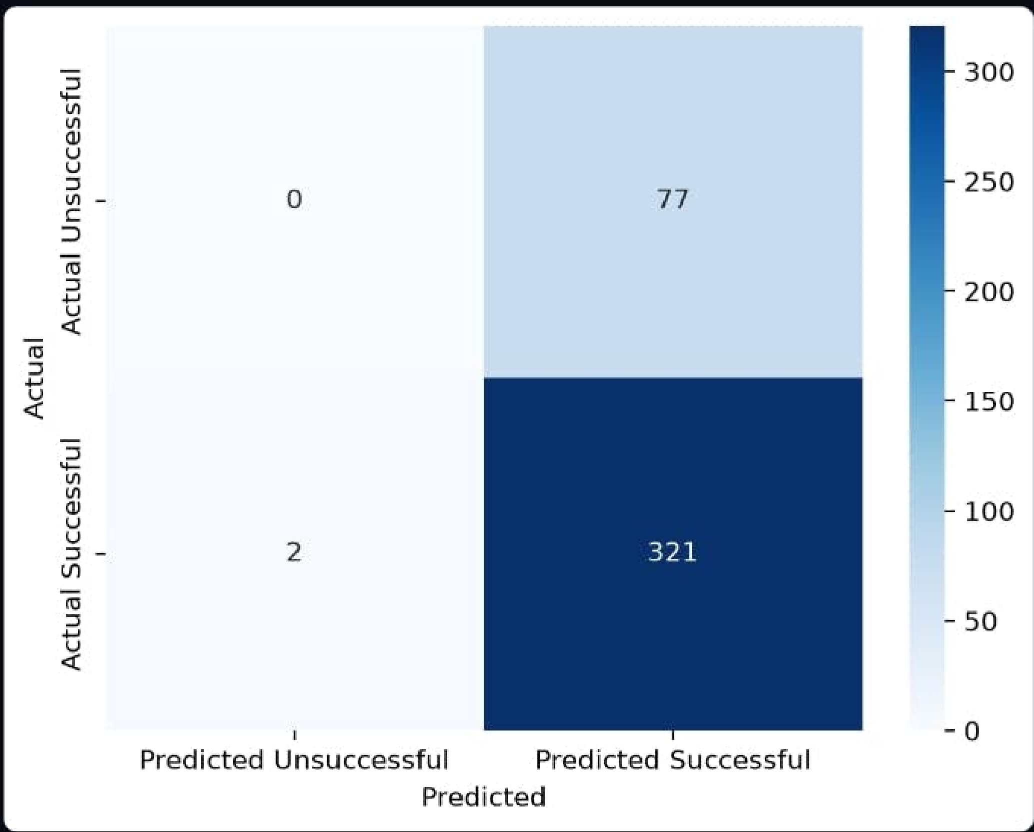
⭐ Minimum Vote Average
0.00

📊 Confusion Matrix Visualization



Statistical tests are used to identify whether relationships and differences observed in the movie dataset are statistically significant.

🧪 T-Test: Popularity of Successful vs Unsuccessful Movies

The independent samples t-test examines whether the average popularity score differs significantly between financially successful and unsuccessful movies.

Null Hypothesis (H_0): The mean popularity of successful and unsuccessful movies is equal.

Alternative Hypothesis (H_1): The mean popularity of successful and unsuccessful movies is different.

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🎛️ Data Filters

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All

⭐ Minimum Vote Average
0.00

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The independent samples t-test examines whether the average popularity score differs significantly between financially successful and unsuccessful movies.

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Alternative Hypothesis (H_1): The mean popularity of successful and unsuccessful movies is different.

T-Statistic	P-Value
2.0617	0.039682

The result is statistically significant at the 5% significance level. We reject the null hypothesis. This suggests that the average popularity differs significantly between successful and unsuccessful movies.

A p-value below 0.05 is considered statistically significant at the 5% significance level.

🧪 Chi-Square Test: Genre vs Movie Success

The Chi-Square test examines whether movie genre and financial success are statistically associated.

Chi-Square Statistic	P-Value	Degrees of Freedom
1.7731	0.994569	9

📋 Genre vs Success Contingency Table

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🎭 Filter by Genre

All

⭐ Minimum Vote Average
0.00

📋 Genre vs Success Contingency Table

genre	0	1
Action	41	151
Adventure	41	166
Animation	38	165
Comedy	38	164
Drama	35	162
Horror	36	163
Romance	42	178
Science Fiction	44	161
Thriller	37	157
Unknown	34	147

The relationship between movie genre and financial success is not statistically significant at the 5% significance level.

💡 Statistical Insights

The t-test helps determine whether successful and unsuccessful movies have significantly different average revenues.

The Chi-Square test helps determine whether a movie's genre is associated with its likelihood of being financially successful.

A p-value below 0.05 is considered statistically significant at the 5% significance level. However, statistical significance does not necessarily imply a strong or causal relationship.

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⭐ Minimum Vote Average

0.00

💵 Revenue Predictor

Predict the potential box office revenue of a movie using Machine Learning.

Enter the movie details below. MovieIQ will use a Linear Regression model trained on historical movie data to estimate potential revenue.

🧠 Prediction Model

Model	📊 Input Features	🎯 Target
Linear ...	4	Movie R...

📈 Model Performance

R^2 Score	Mean Absolute Error
0.5981	$65,621,488

R^2 Score indicates how well the model explains variations in movie revenue. Mean Absolute Error represents the average difference between the predicted and actual revenue.

🎬 Enter Movie Details

💰 Production Budget ($)

100000000

⏱️ Runtime (Minutes)

120

🔥 Popularity Score

50.00

⭐ Expected Vote Average

6.00

🎬 Predict Box Office Revenue

Statistical tests are used to identify whether relationships and differences observed in the movie dataset are statistically significant.

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⭐ Minimum Vote Average
0.00

📊 Machine Learning Model Comparison

MovieIQ uses different Machine Learning models to answer different business questions.

Random Forest Classification predicts whether a movie is financially successful, while regression models predict the expected box office revenue.

Model Performance

🌲 Random Forest

Classification ...

80.2...

Predicts whether a movie is financially successful.

📈 Linear Regression

R² Score

0.5981

Mean Absolute...

$65,...

Predicts expected box office revenue.

🌲 Random Forest Regression

R² Score

0.5359

Mean Absolute...

$69,...

Predicts expected box office revenue.

📊 Classification Metrics

Precision

80.6...

Recall

99.3...

📋 Model Comparison

	Model	Problem Type	Prediction	Primary Metric	Performanc
0	Random Forest Classification	Classification	Movie Success	Accuracy	80.25%
1	Linear Regression	Regression	Movie Revenue	R^2 Score	0.5981
2	Random Forest Regression	Regression	Movie Revenue	R^2 Score	0.5359

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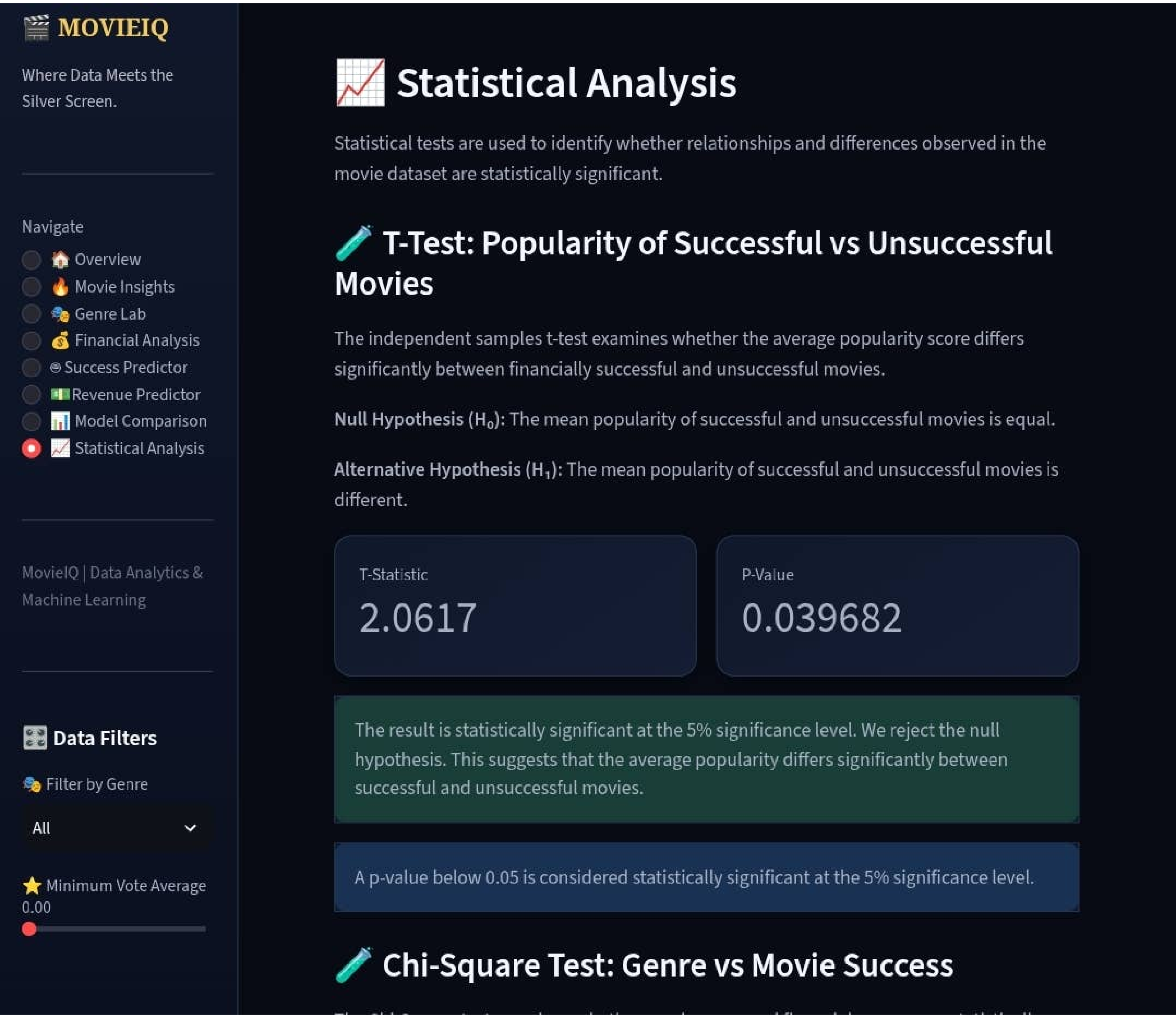
💡 Model Selection Insight

The models serve different purposes. Random Forest Classification predicts whether a movie is likely to be financially successful, while Linear Regression and Random Forest Regression estimate the actual revenue.

> For movie success prediction, Random Forest Classification is evaluated using Accuracy. For revenue prediction, regression models are evaluated using R^2 Score and Mean Absolute Error.

Statistical tests are used to identify whether relationships and differences observed in the movie dataset are statistically significant.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0.00
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T-Statistic
2.0617
P-Value
0.039682
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🎬 About MovieIQ

MovieIQ is a movie analytics and revenue prediction dashboard that combines data analysis, business intelligence, and machine learning to explore movie performance and box office trends.

Built with Python • Pandas • Scikit-learn • Streamlit

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